

Install Java, Tomcat, and Eclipse

Installing Java and Tomcat

1. `# dnf install java-17-openjdk-headless* -y`
2. `# java -version` (**take a screenshot**)
3. `# useradd -r -m -U -d /opt/tomcatserver -s /bin/false tomcatuser`
4. `# wget https://archive.apache.org/dist/tomcat/tomcat-10/v10.1.17/bin/apache-tomcat-10.1.17.tar.gz` //downloading tomcat tar file.
5. `# tar -xvf apache-tomcat-10.1.17.tar.gz -C /opt/tomcatserver`
6. `# ln -s /opt/tomcatserver/apache-tomcat-10-1-17 /opt/tomcatserver/updated`
7. `# chown -R tomcatuser: /opt/tomcatserver/*`
8. `# sh -c 'chmod +x /opt/tomcatserver/updated/bin/*.sh'`
9. `# nano /etc/systemd/system/tomcat.service`

Type the following:

[Unit]

Description=Apache Tomcat Web Application by [type your full name]

After=network.target

[Service]

Type=forking

Environment="JAVA_HOME=/usr/lib/jvm/java-17-openjdk-17.0.11.0.9-2.el9.x86_64"

Environment="CATALINA_PID=/opt/tomcatserver/updated/temp/tomcat.pid"

Environment="CATALINA_HOME=/opt/tomcatserver/updated/"

Environment="CATALINA_BASE=/opt/tomcatserver/updated/"

Environment="CATALINA_OPTS=-Xms512M -Xmx1024M -server -XX:+UseParallelGC"

Environment='JAVA_OPTS=-Djava.awt.headless=true'

ExecStart=/opt/tomcatserver/updated/bin/startup.sh

ExecStop=/opt/tomcatserver/updated/bin/shutdown.sh

User=tomcatuser

Group=tomcatuser

[Install]

WantedBy=multi-user.target

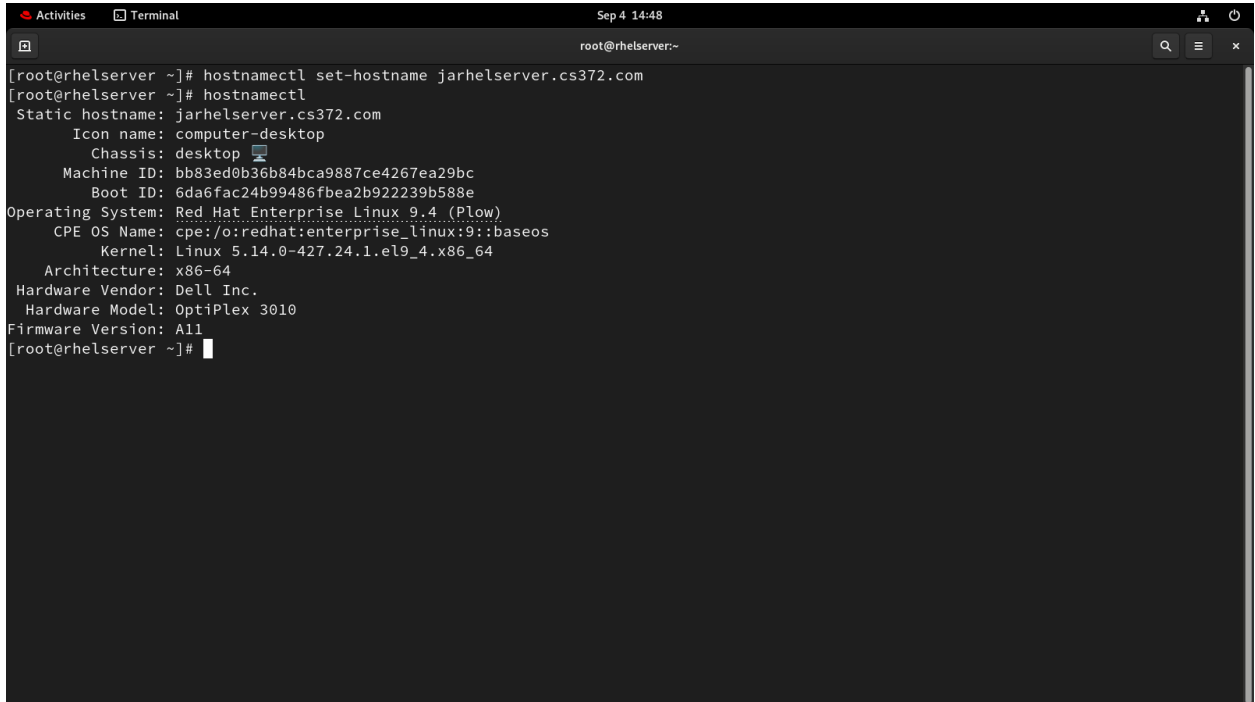
Save the file by pressing: *Ctrl +o* and **exit nano** by pressing: *Ctrl+x*

10. `# systemctl daemon-reload` //to update the system daemon.
11. `# systemctl start tomcat` // start the tomcat service

Assignment submission

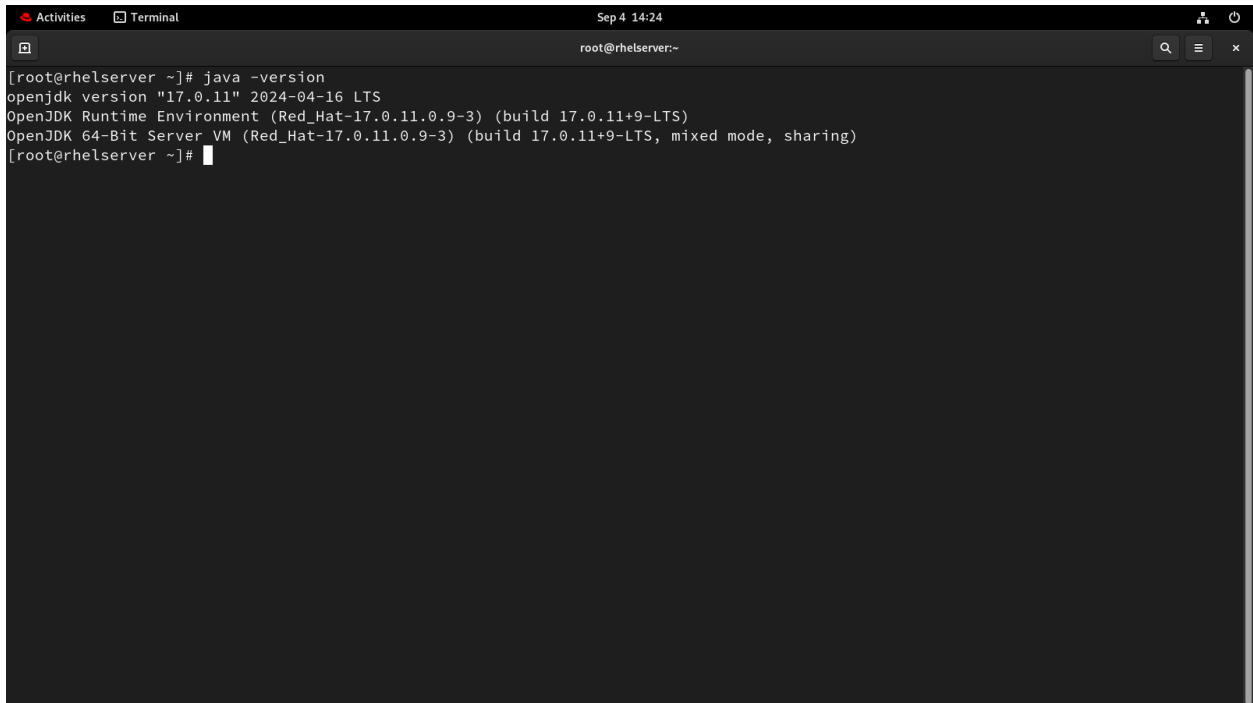
Perform the following steps to complete the assignment

1. Launch MS. Word and open a blank document (5 points)
2. Open your Linux terminal (5 points)
3. Change your Linux hostname (*hostnamectl set-hostname yourinitialrhelserver.cs372.com* (ex: jarhelserver.cs372.com, where ja=Jean Alcime)
Your result should look similar to the one below (15 points if successful)



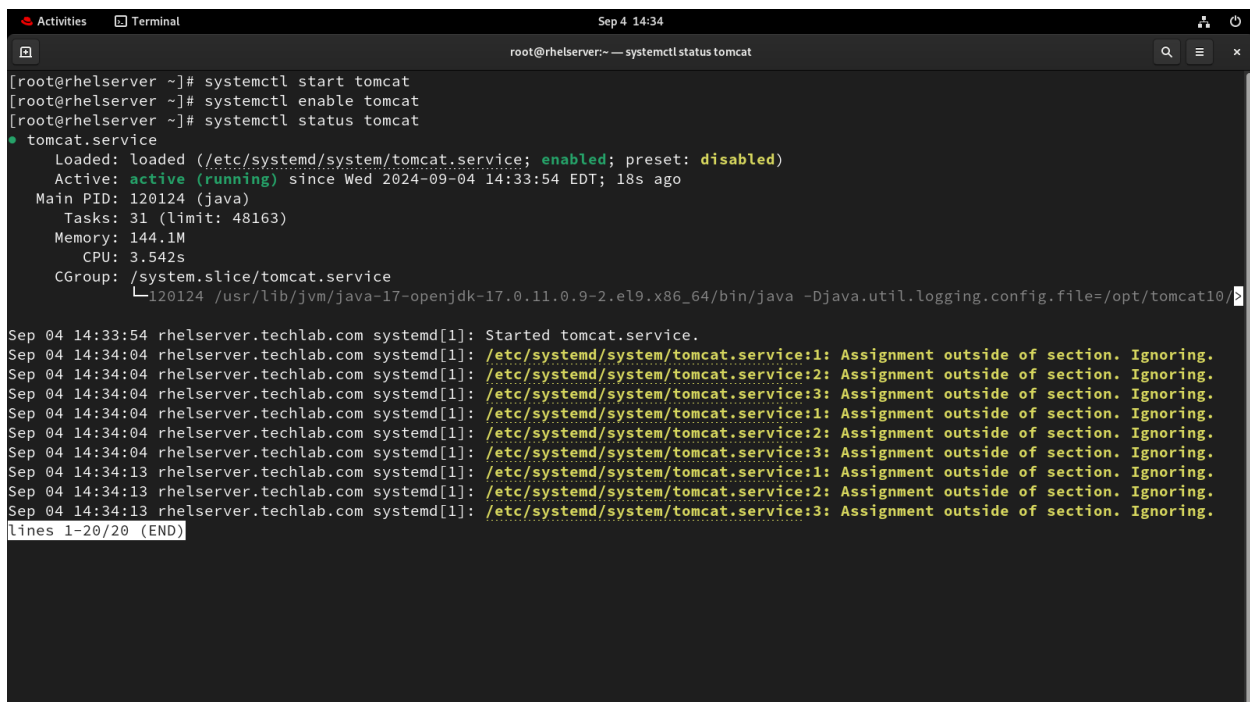
```
Activities Terminal Sep 4 14:48 root@rhelserver-~
[root@rhelserver ~]# hostnamectl set-hostname jarhelserver.cs372.com
[root@rhelserver ~]# hostnamectl
Static hostname: jarhelserver.cs372.com
Icon name: computer-desktop
Chassis: desktop
Machine ID: bb83ed0b36b84bca9887ce4267ea29bc
Boot ID: 6da6fac24b99486fbea2b922239b588e
Operating System: Red Hat Enterprise Linux 9.4 (Plow)
CPE OS Name: cpe:/o:redhat:enterprise_linux:9::baseos
Kernel: Linux 5.14.0-427.24.1.el9_4.x86_64
Architecture: x86-64
Hardware Vendor: Dell Inc.
Hardware Model: OptiPlex 3010
Firmware Version: A11
[root@rhelserver ~]#
```

4. At the terminal, type: *java -version* and press Enter (5 points)
5. Take a screenshot of the result (15 points for successful result), the result should look similar to the one below.

A terminal window titled 'Sep 4 14:24' with the prompt 'root@rhelserver:~'. The user has entered the command 'java -version'. The output shows: 'openjdk version "17.0.11" 2024-04-16 LTS', 'OpenJDK Runtime Environment (Red_Hat-17.0.11.0.9-3) (build 17.0.11+9-LTS)', and 'OpenJDK 64-Bit Server VM (Red_Hat-17.0.11.0.9-3) (build 17.0.11+9-LTS, mixed mode, sharing)'. The prompt is now '[root@rhelserver ~]#'.

```
[root@rhelserver ~]# java -version
openjdk version "17.0.11" 2024-04-16 LTS
OpenJDK Runtime Environment (Red_Hat-17.0.11.0.9-3) (build 17.0.11+9-LTS)
OpenJDK 64-Bit Server VM (Red_Hat-17.0.11.0.9-3) (build 17.0.11+9-LTS, mixed mode, sharing)
[root@rhelserver ~]#
```

6. Type: *clear* and press Enter (5 points)
7. Type: *systemctl start tomcat* and press Enter (5 points)
8. Type: *systemctl enable tomcat* (5 points)
9. Type: *systemctl status tomcat* (5 points)
10. Take a screenshot of the result, your result should look similar to the one below. (25 points if successful—**active (running)**)

A terminal window titled 'Sep 4 14:34' with the prompt 'root@rhelserver:~ -- systemctl status tomcat'. The user has entered three commands: 'systemctl start tomcat', 'systemctl enable tomcat', and 'systemctl status tomcat'. The output for 'systemctl status tomcat' shows the service is 'loaded' and 'active (running)'. It also displays system logs for the service start. The prompt is now '[root@rhelserver ~]#'.

```
[root@rhelserver ~]# systemctl start tomcat
[root@rhelserver ~]# systemctl enable tomcat
[root@rhelserver ~]# systemctl status tomcat
• tomcat.service
   Loaded: loaded (/etc/systemd/system/tomcat.service; enabled; preset: disabled)
   Active: active (running) since Wed 2024-09-04 14:33:54 EDT; 18s ago
     Main PID: 120124 (java)
       Tasks: 31 (limit: 48163)
      Memory: 144.1M
         CPU: 3.542s
    CGroup: /system.slice/tomcat.service
            └─120124 /usr/lib/jvm/java-17-openjdk-17.0.11.0.9-2.el9.x86_64/bin/java -Djava.util.logging.config.file=/opt/tomcat10/

Sep 04 14:33:54 rhelserver.techlab.com systemd[1]: Started tomcat.service.
Sep 04 14:34:04 rhelserver.techlab.com systemd[1]: /etc/systemd/system/tomcat.service:1: Assignment outside of section. Ignoring.
Sep 04 14:34:04 rhelserver.techlab.com systemd[1]: /etc/systemd/system/tomcat.service:2: Assignment outside of section. Ignoring.
Sep 04 14:34:04 rhelserver.techlab.com systemd[1]: /etc/systemd/system/tomcat.service:3: Assignment outside of section. Ignoring.
Sep 04 14:34:04 rhelserver.techlab.com systemd[1]: /etc/systemd/system/tomcat.service:1: Assignment outside of section. Ignoring.
Sep 04 14:34:04 rhelserver.techlab.com systemd[1]: /etc/systemd/system/tomcat.service:2: Assignment outside of section. Ignoring.
Sep 04 14:34:04 rhelserver.techlab.com systemd[1]: /etc/systemd/system/tomcat.service:3: Assignment outside of section. Ignoring.
Sep 04 14:34:13 rhelserver.techlab.com systemd[1]: /etc/systemd/system/tomcat.service:1: Assignment outside of section. Ignoring.
Sep 04 14:34:13 rhelserver.techlab.com systemd[1]: /etc/systemd/system/tomcat.service:2: Assignment outside of section. Ignoring.
Sep 04 14:34:13 rhelserver.techlab.com systemd[1]: /etc/systemd/system/tomcat.service:3: Assignment outside of section. Ignoring.
lines 1-20/20 (END)
```

11. Save the file as follow: yourlastname_firstinitial_cs372_RHEL_Install
(*Alcime_j_cs372_RHEL_installed*): 10 points.

Total Points: 100